Recitation 01 & 02
Data Types & Strings

Question 1 | Evaluate expression

1) Using the Python interpreter, type *1 + 2* and then hit return. Check out what happens.
2) * is the multiplication operator, and ** is the exponentiation operator. Experiment by entering different expressions and recording what is displayed by the Python interpreter.

Question 2 | Syntax error

1) Type *1 2* and then hit return. Check out what happens this time.

In many cases, Python indicates where the syntax error occurred, but it is not always right, and it doesn't give you much information about what is wrong. So, for the most part, the burden is on you to learn the syntax rules.

Question 3 | Print hello

1) Type *print("hello")* at the Python prompt. Check out what happens.
2) Now type *"hello"* and check your result.

Question 4 | Say "cheese"

1) Type *cheese* without the quotation marks. What kind of error is this?

Question 5 | Shell vs. script mode

1) Type *6 + 4 * 9* in the Python prompt and hit enter. Check what happens.
2) Now create a python script named *test1.py* with the following contents:

   ```
   6+4*9
   ```

   What happens when you run this script?
3) Now change the contents to:

   ```
   print(6 + 4 * 9)
   ```

   What happens when you run the script?

Question 6 | Say "cheese"

1) Write a program named *using_print1.py* that displays the following then run from the command prompt:

```
This is line 1.
This is line 2, with a blank line above it.
```

Question 7 | Binary Numbering System

Numbers are represented in decimal system by default in Python. But it is also possible to input positive integers in binary directly.

1) Type in the interpreter *0b10011010* and check out what happens.
2) Try different binary values and check if the conversion is correct.

Python also has a built-in function that allows one to display numbers in binary representation.

3) Type *bin(100)* and check out what happens. What kind of output is displayed?
4) Try with different values.

Question 8 | Hexadecimal Numbering System

In Python, you can input positive integers in hexadecimal directly.

1) Type in the interpreter *0xAB* and check out what happens.
2) Try different hexadecimal values and check if the conversion is correct.

Python also has a built-in function that allows you to display numbers in hexadecimal representation.

3) Type *hex(100)* and check out what happens. What kind of output is displayed?
4) Fill the following table by converting the values provided in one of the columns to corresponding missing representation. Use Python's built-in functions to check if the conversions are correct.

Decimal	**Binary**	**Hexadecimal**
	0b1100	
100		
		0xA4
247		
		0x12C

Question 9 | Numeric values and operations

Reduce the following expressions to their corresponding numerical values, and specify their type.

5 * -6		34 / 10	
34 // 10		34 % 10	
23 % 4.0		25 + 30 / 6	
(1 + 3) * (6 / 2)		(1 + 3.0) * (6 // 2)	
(21 % 12) % 7		100 - 25 * 3 % 4	
3 * (17 / 3) + 17 % 3		(21 % 7) % 12	

Question 10 | Strings

Write out the output of each line in the space provided. If the line results in an error, write "error".

print(5 + “5”) print(“5”+”5”)		print(5 + int(“5”))	
print(str(5) + int("5"))		print(15 % 2) print(str(14%7))	
print("hey" + "yo")		print(4 * “Let it be”)	
print("yo" * 4.0) print(“yo”*int(4.0))		print(“hi” + there) there = “there” print(“hi”+there)	
print("You talkin’ to me? " * 3 + "Then who the hell else are you talking to?")			
print(“Let it be” * 2 + "Let it be ” * 2)			
ord('a')		chr(97+25)	
str(97+25)		len('97'+'25')	

Question 11 | Escape sequence

print("My name is \'Promethee\'")	
print("My name is "Promethee")	
print('My name is "Promethee') print('My name is \”Promethee')	
print("My name is \"Promethee\"")	
print("My first name is\t\t\"Promethee\"\nMy last name is\t\t\”Spathis\"")	
print("My first name is\t\t\"Promethee\"\rMy last name is\t\t\"Spathis\"")	

Question 12 | String indexing & slicing

`"HellO WorlD\n"[-1]`	
`"HellO WorlD\n"[-2]`	
`len("HellO WorlD\n")`	
`"HellO WorlD\n"[-len("HellO WorlD\n")]`	
`"HellO WorlD\n"[-13]`	
`"HellO WorlD\n"[:5]`	
`"HellO WorlD\n"[6:500]`	
`"HellO WorlD\n"[7:]`	
`"HellO WorlD\n"[:7] + "HellO WorlD\n"[7:]`	
`"HellO WorlD\n"[5:6]`	
`"HellO WorlD\n"[6:5]`	
`"HellO WorlD\n"[-6:-1]`	
`"HellO WorlD\n"[:5:2]`	
`"HellO WorlD\n"[5::-1]`	
`"HellO WorlD\n"[-2:-7:-1]`	